

PROJECT INTRODUCTION & OVERVIEW

Data-Driven Pharmaceutical Sales Planning & Doctor Targeting Model

A Multi-Attribute Decision Model for Resource Allocation under Constraint

1 What this project is

This project builds a **decision model** that answers a single practical question: when a pharmaceutical company has far more doctors than it can possibly visit, which doctors should it invest in, how much should it invest in each, and how can that choice be justified with evidence rather than intuition?

The setting is a regional launch of a new medicine. The company has roughly **600 relevant prescribing doctors** in its territory, **six medical representatives (MRs)** to visit them, a promotional budget of **₹30 lakhs per quarter**, and about **six months** in which the launch will succeed or fail. These four constraints operate at the same time, and together they make the conventional approach — try to cover everyone — arithmetically impossible rather than merely inefficient.

The project produces three things: a **scoring model** that ranks every doctor on five weighted dimensions; a **resource-allocation system** that converts those scores into visit schedules and budget entitlements; and a **working software tool** that implements both so the model can actually be used rather than only described.

2 Why this topic matters

Pharmaceutical selling is unusual: the company does not sell to the person who consumes the product. It persuades a doctor to prescribe, and the doctor decides on clinical grounds. Influence therefore runs through scientific credibility and relationships built over repeated contact — which makes the medical representative's time the scarcest and most valuable resource the company controls.

That scarcity is the heart of the problem. Consider the arithmetic:

Available calls = 6 MRs × 22 working days × 9 calls/day = 1,188 per month
1,188 calls ÷ 600 doctors ≈ one visit per doctor every two months

Even before reserving time for chemists, hospitals and administration

A doctor visited once every two months forms no prescribing habit. Splitting the budget equally gives ₹5,000 per doctor per quarter — too little to fund meaningful scientific engagement for anyone. Uniform coverage therefore produces **uniform ineffectiveness**. The company must choose, and the only real question is whether it chooses deliberately, using a defensible method, or by accident.

THE QUESTION THE PROJECT REFRAMES

The conventional question is “how do we reach all 600 doctors?” — a question whose honest answer is “we cannot”. This project replaces it with: **“which doctors will generate the majority of prescriptions, how do we identify them before the launch rather than after, and how do we prove the resulting plan is deliverable?”**

3 The central idea: the Doctor Value Score

Traditional targeting ranks doctors on one axis — patient volume, usually as prescribing deciles. This is convenient and dangerously incomplete. A doctor seeing 50 relevant patients a day who is loyal to a competitor and has adopted no new molecule in four years is worth less than one seeing 20 patients who adopts early, teaches at a medical college, and is dissatisfied with existing options. A single-axis model cannot express that; a multi-attribute model can.

The project therefore scores every doctor on five dimensions and combines them into one **Doctor Value Score (DVS)** on a 0–10 scale:

$$DVS = (P_1 \times 0.30) + (P_2 \times 0.25) + (P_3 \times 0.20) + (P_4 \times 0.15) + (P_5 \times 0.10)$$

The five pillars, weighted by how directly each determines realised prescriptions

#	Pillar	Weight	What it captures
P ₁	Prescription Potential	30%	Relevant patients per month — the only pillar mapping directly to prescriptions
P ₂	Adoption Propensity	25%	Willingness to try a new molecule within the launch window
P ₃	Influence & Network Power	20%	Whether the doctor shapes other doctors’ prescribing
P ₄	Accessibility	15%	Whether the doctor can realistically be met
P ₅	Competitive Vulnerability	10%	Whether a competitor’s hold can be broken

Doctors are then banded into four tiers — **Platinum** (DVS 8.0–10.0), **Gold** (6.0–7.9), **Silver** (4.0–5.9) and **Bronze** (below 4.0) — and each tier receives a different visit frequency and budget. The top tier receives roughly **4.15 times** the spend per doctor that uniform allocation would give, funded entirely by withdrawing effort from doctors unlikely to prescribe.

The Doctor Targeting Pyramid — resources concentrate toward the apex

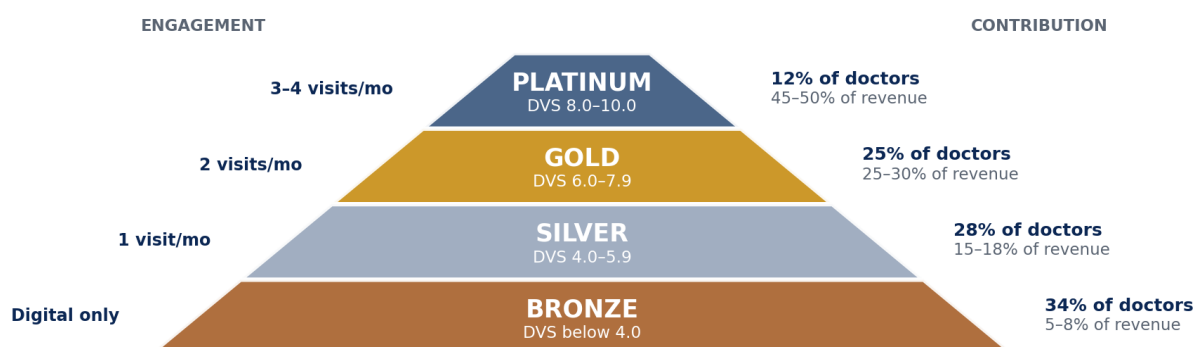


Figure 1 — The targeting pyramid. Engagement intensity rises toward the apex while population share falls, producing the concentration on which the strategy depends.

4 The academic foundation

The project's central claim is that none of its formulas are invented for convenience. Each rests on established theory, and the references below are hyperlinked so any claim can be checked at source.

4.1 Why the scores are added

The most obvious objection to the DVS is: why add weighted scores rather than multiply them, or take the minimum? The answer is that the weighted additive form is the canonical solution in **Multi-Attribute Utility Theory**, formalised by **Keeney & Raiffa (1976)** [[theory overview](#)]. Their Theorem 3.6 establishes that such a function exists precisely when the attributes are mutually preferentially independent — a specific, checkable condition, which the project states openly rather than assuming silently.

The practical form used here — rate each attribute on a common anchored scale, weight, sum — is the **SMART** method of **Edwards (1977)** [[doi:10.1109/TSMC.1977.4309720](#)]. Edwards argued that elaborate elicitation introduces more error than it removes, because complex hypothetical questions produce unreliable answers. For a model applied by representatives across 600 doctors, that argument is decisive. The caution that different weighting methods yield materially different weights comes from **Pöyhönen & Hämäläinen (2001)** [[doi:10.1016/S0377-2217\(99\)00467-1](#)], and the recommended upgrade path is the **Analytic Hierarchy Process** [[AHP overview](#)].

4.2 Why higher-value doctors get more visits

This is the project's strongest analytical result. Empirical work shows that response to detailing is **concave** — the second call in a month produces less additional prescribing than the first, the third less than the second. This is the foundation of Lodish's CALLPLAN system [[doi:10.1287/mnsc.18.4.P25](#)] and is confirmed in modern econometric work by **Manchanda, Rossi & Chintagunta (2004)** [[doi:10.1509/jmkr.41.4.467.47005](#)].

Under concavity with a binding capacity constraint, maximising total response requires equalising the **marginal** return of the last call across doctors — not the total return. Doctors who respond more strongly should therefore receive more calls, until their marginal return falls to a common level. A four-tier frequency ladder is a step approximation of that continuous optimum. This converts what most strategy documents assert as management preference into a derived conclusion.

A FINDING WORTH NOTING

Manchanda et al. also found that in practice, pharmaceutical companies detail **high-volume physicians more heavily regardless of how responsive those physicians actually are** — exactly the error a multi-dimensional score is designed to prevent. The literature does not merely support this project's approach; it documents the failure the approach corrects.

4.3 Why adoption speed is scored separately

Pillar 2 rests on diffusion theory. **Rogers** classifies adopters as Innovators (2.5%), Early Adopters (13.5%), Early Majority (34%), Late Majority (34%) and Laggards (16%) [[diffusion of innovations](#)]. In a six-month launch, a medium-volume early adopter is worth more than a high-volume laggard, because the laggard will not prescribe within the window whatever effort is spent.

The formal counterpart is the **Bass diffusion model** [[doi:10.1287/mnsc.15.5.215](#)], which shows cumulative adoption follows an S-curve driven by innovation (p) and imitation (q) coefficients. Benchmark values of $p \approx 0.03$ and $q \approx 0.38$ come from a meta-analysis of 213 applications

[[doi:10.1177/002224379002700107](https://doi.org/10.1177/002224379002700107)]. The project uses this to test whether its own forecast is realistic — and finds the plan’s linear ramp is optimistic in the early months.

4.4 Why influence is a separate dimension — with a caveat

Pillar 3 rests on the classic **Columbia tetracycline study** (Coleman, Katz & Menzel, 1966), which found that network-central doctors adopted earlier and that their adoption preceded acceleration of the diffusion curve. It also supplies a usable field definition of an opinion leader: three or more peer nominations, rather than job title.

THE CAVEAT THAT MAKES THIS HONEST

Van den Bulte & Lilien (2001) [[doi:10.1086/320819](https://doi.org/10.1086/320819)] re-analysed that very dataset with newly collected advertising data and found the peer-influence effect **largely disappears once marketing effort is controlled for** — the company had simply detailed influential doctors harder. This is a textbook case of correlation mistaken for causation, and this project therefore treats opinion-leader influence as real but bounded, using the econometrically identified estimates of **Nair, Manchanda & Bhatia (2010)** [[doi:10.1509/jmkr.47.5.883](https://doi.org/10.1509/jmkr.47.5.883)] rather than the inflated multipliers common in industry practice.

4.5 Territory design and concentration measurement

Assigning doctors to representatives so that workload is balanced — not headcount — is formally **makespan minimisation on identical parallel machines**, an NP-hard scheduling problem. The project uses the Longest Processing Time heuristic, which **Graham (1969)** [[doi:10.1137/0117039](https://doi.org/10.1137/0117039)] proved is guaranteed to come within $(4/3 - 1/3m)$ of optimal — about 1.28× for six territories. That real territory design is a two-criteria problem, balancing workload against travel, comes from **Zoltners & Sinha (2005)** [[doi:10.1287/mksc.1050.0133](https://doi.org/10.1287/mksc.1050.0133)].

Finally, the project tests its own premise. The concentration of value is measured with the **Gini coefficient** [[Overview: Gini coefficient](#)] and a Lorenz curve, giving a quantitative version of the **Pareto principle** [[Overview: Pareto principle](#)] the strategy invokes. If value turned out to be evenly spread, differentiated targeting would be unjustified and the whole approach would fail — which is precisely why the test belongs in the analysis.

4.6 The regulatory boundary

Every rupee in the model is constrained by the **Uniform Code for Pharmaceutical Marketing Practices (UCPMP) 2024**, notified by India’s Department of Pharmaceuticals on 12 March 2024 [[published summary \(PDF\)](#); [issuing department](#)]. It prohibits gifts, monetary grants, travel and hospitality to healthcare professionals; caps free samples at 12 packs per drug per professional per year with total value not exceeding 2% of prior-year domestic sales; and caps brand reminders at ₹1,000. The per-doctor budget in this model is therefore an **entitlement ceiling for compliant scientific activity** — samples, CME and medical education — not a transfer of value to the doctor.

5 How the project was carried out

Stage	What was done
1. Problem analysis	Quantified the constraint set and demonstrated arithmetically that uniform coverage cannot work.
2. Model design	Defined five pillars with anchored 0–10 rubrics, assigned weights, and set tier thresholds.
3. Formula derivation	Derived ten formulas from first principles, each with its theoretical authority, assumptions, limitations and a worked example.
4. Verification	Independently recomputed all 123 numerical assertions in the source strategy using a Python audit harness.
5. Implementation	Built DVS Manager, a browser-based tool implementing every formula, verified by 70 automated tests.
6. Critical evaluation	Documented where the model is weakest and what would be needed to validate it empirically.

The verification stage proved the most valuable. Of 123 assertions, **85 verified exactly**, but **six material errors** emerged that were invisible on a careful reading and only appeared on recomputation — including a breakeven date overstated by five to six months, and a headline “140% ROI” that was in fact a revenue-to-cost ratio (the true figure is 40%, or roughly –2% once gross margin is applied). This is the strongest practical argument for building strategy as an auditable model: **the arithmetic checks the reasoning**.

6 What the project delivers

Outcome	Result	Where it is derived
Doctors identified as high-value	200 of 600 (~33%), expected to drive ~75% of revenue	Formulas 1–2
Spend concentration on top tier	4.15× the uniform allocation	Formula 5
Visit frequency, top vs bottom tier	3.5 vs 0.33 calls per month	Formula 4
Field-force feasibility	63% utilisation — deliverable with headroom	Formula 3
Territory workload balance	Coefficient of variation 1.8% (tolerance 10%)	Formula 6
Assertions independently verified	85 of 123 exact; 6 material errors corrected	Audit harness

SCOPE AND HONEST LIMITATIONS

This is a **decision-support model, not a validated predictive one**. The five weights are a structured expert prior, not coefficients fitted to prescription data — because at launch no such data exists for a new molecule. The visit frequencies are derived from the shape of the response curve, not from measured elasticities. Both should be re-estimated empirically after two to three quarters of real prescribing data. The model is built to make those assumptions visible and testable rather than to hide them.

7 How to read the full report

Sections	Content
1 – 4	The business problem, strategic framework, and all five pillars in detail
5 – 10	The analytical core. Ten formulas, each derived with proof, assumptions, limitations and a verified worked example
11	Engagement playbooks, KPIs and the quarterly recalibration cycle
12	Critical evaluation — the six errors found and what they mean
13	Conclusion: what is established and what remains to be proved
References	23 sources, each mapped to the formula it supports
Appendix A	One-page formula reference card

COMPANION DELIVERABLES

Pharma_Sales_Strategy_Report.pdf — the full 41-page report with all derivations.

dvs_manager.html — the working tool: enter doctors and representatives, score them, and watch tiering, budget and field-force feasibility recompute instantly. Runs offline in any browser with no installation.

Sources cited in this introduction

All links were verified against the Crossref registry at build time; titles and years match the citations used here. DOI links are permanent identifiers and resolve to the publisher's page.

1. Keeney, R. L. & Raiffa, H. (1976). Decisions with Multiple Objectives: Preferences and Value Tradeoffs. New York: Wiley. [[Overview: Multi-attribute utility theory](#)]

USED FOR: The additive value function and the preferential-independence condition behind the DVS.

2. Edwards, W. (1977). "How to use multiattribute utility measurement for social decision making." IEEE Transactions on Systems, Man, and Cybernetics, 7(5), 326–340. [[doi:10.1109/TSMC.1977.4309720](#)]

USED FOR: SMART — the anchored-scale, direct-weight method the scoring rubric uses.

3. Pöyhönen, M. & Hämäläinen, R. P. (2001). "On the convergence of multiattribute weighting methods." European Journal of Operational Research, 129(3), 569–585. [[doi:10.1016/S0377-2217\(99\)00467-1](#)]

USED FOR: Evidence that elicitation methods disagree — why the weights are treated as a prior.

4. Saaty, T. L. (1980). The Analytic Hierarchy Process. New York: McGraw-Hill. [[Overview: Analytic Hierarchy Process](#)]

USED FOR: The recommended upgrade path for deriving weights with a consistency check.

5. Lodish, L. M. (1971). "CALLPLAN: An interactive salesman's call planning system." Management Science, 18(4-II), 25–40. [[doi:10.1287/mnsc.18.4.P25](#)]

USED FOR: Origin of concave sales response and optimised call allocation.

6. Manchanda, P., Rossi, P. E. & Chintagunta, P. K. (2004). "Response Modeling with Nonrandom Marketing-Mix Variables." Journal of Marketing Research, 41(4), 467–478. [[doi:10.1509/jmkr.41.4.467.47005](#)]

USED FOR: Modern confirmation of concavity; documents that firms over-detail high-volume, low-response physicians.

7. Bass, F. M. (1969). "A New Product Growth for Model Consumer Durables." Management Science, 15(5), 215–227. [[doi:10.1287/mnsc.15.5.215](#)]

USED FOR: The S-curve adoption model used to test the realism of the prescriber ramp.

8. Rogers, E. M. (2003). Diffusion of Innovations, 5th ed. New York: Free Press. [[Overview: Diffusion of innovations](#)]

USED FOR: Adopter categories underpinning Pillar 2, Adoption Propensity.

9. Sultan, F., Farley, J. U. & Lehmann, D. R. (1990). "A Meta-Analysis of Applications of Diffusion Models." Journal of Marketing Research, 27(1), 70–77. [[doi:10.1177/002224379002700107](#)]

USED FOR: Benchmark diffusion parameters $p \approx 0.03$ and $q \approx 0.38$ across 213 applications.

10. Coleman, J. S., Katz, E. & Menzel, H. (1966). Medical Innovation: A Diffusion Study. Indianapolis: Bobbs-Merrill.

USED FOR: The Columbia tetracycline study — empirical basis for Pillar 3, Influence.

11. Van den Bulte, C. & Lilien, G. L. (2001). "Medical Innovation Revisited: Social Contagion versus Marketing Effort." American Journal of Sociology, 106(5), 1409–1435. [[doi:10.1086/320819](#)]

USED FOR: The essential corrective: contagion effects fade once marketing effort is controlled for.

12. Nair, H. S., Manchanda, P. & Bhatia, T. (2010). "Asymmetric Social Interactions in Physician Prescription Behavior: The Role of Opinion Leaders." Journal of Marketing Research, 47(5), 883–895. [[doi:10.1509/jmkr.47.5.883](#)]

USED FOR: A credible, econometrically identified size for opinion-leader influence.

13. Graham, R. L. (1969). "Bounds on Multiprocessing Timing Anomalies." SIAM Journal on Applied Mathematics, 17(2), 416–429. [[doi:10.1137/0117039](https://doi.org/10.1137/0117039)]

USED FOR: The LPT worst-case guarantee used for territory workload balancing.

14. Zoltners, A. A. & Sinha, P. (2005). "Sales Territory Design: Thirty Years of Modeling and Implementation." Marketing Science, 24(3), 313–331. [[doi:10.1287/mksc.1050.0133](https://doi.org/10.1287/mksc.1050.0133)]

USED FOR: Territory design as a two-criteria optimisation of workload and travel.

15. Gini, C. (1912) / Lorenz, M. O. (1905). Concentration measurement. [[Overview: Gini coefficient](#)]

USED FOR: The Gini coefficient and Lorenz curve used to test whether targeting is warranted.

16. Pareto, V. (1896). Cours d'Économie Politique. Lausanne: F. Rouge. [[Overview: Pareto principle](#)]

USED FOR: The concentration principle the targeting strategy operationalises.

17. Department of Pharmaceuticals, Government of India (2024). Uniform Code for Pharmaceutical Marketing Practices (UCPMP) 2024, notified 12 March 2024. [[National Medical Journal of India summary \(PDF\)](#)]

USED FOR: The binding compliance constraint on all promotional spend in the model.

A NOTE ON THE LINKS

DOI (Digital Object Identifier) links are permanent and will not break. Some publisher sites require an institutional login for the full text; the abstract and citation details are always freely visible. Where a paywalled source has a reliable open summary, that has been linked instead so the underlying concept can be checked without a subscription.